Clustering Evaluation Metrics

CA Assignment · Questions 2 & 3

Complete Worked Solutions — Confusion Matrix, Macro-Averaged & B-CUBED

1. Setup: Reading Figure 1

We assign each cluster a majority-class label by inspecting Figure 1:

- **Cluster 1** (leftmost circle): 4 Red, 3 Blue, 3 Purple ⇒ labelled **Red**. Total = 10.
- **Cluster 2** (middle circle): 2 Red, 5 Blue, 2 Purple ⇒ labelled **Blue**. Total = 9.
- **Cluster 3** (rightmost circle): 2 Red, 2 Blue, 4 Purple ⇒ labelled **Purple**. Total = 8.

True class totals: Red = 8, Blue = 10, Purple = 9. Grand total $N = 27$.

2. Question 2 (20 pts): Confusion Matrix & Macro-Averaged Metrics

2.1 Confusion Matrix

Rows are *true* class labels; columns are *predicted* cluster labels (i.e. the cluster a point was assigned to).

True \ Predicted	Red	Blue	Purple	Row total
Red	4	2	2	8
Blue	3	5	2	10
Purple	3	2	4	9
Col total	10	9	8	27

2.2 Per-Cluster Precision

Precision measures how *pure* a cluster is — what fraction of its items belong to the majority (assigned) class:

$$P(\text{cluster}_k) = \frac{\text{count of majority class in cluster}_k}{|\text{cluster}_k|}$$

$$P(\text{Red cluster}) = \frac{4}{10} = 0.4000$$
$$P(\text{Blue cluster}) = \frac{5}{9} \approx 0.5556$$
$$P(\text{Purple cluster}) = \frac{4}{8} = \frac{1}{2} = 0.5000$$

Macro-Averaged Precision (arithmetic mean over the three clusters):

$$\begin{aligned}
\overline{P} &= \frac{1}{3}\left(\frac{4}{10} + \frac{5}{9} + \frac{4}{8}\right) \\
&= \frac{1}{3}\left(\frac{36}{90} + \frac{50}{90} + \frac{45}{90}\right) \\
&= \frac{1}{3} \cdot \frac{131}{90} = \frac{131}{270}
\end{aligned}$$

$$\overline{P} = \frac{131}{270} \approx 0.4852$$

2.3 Per-Class Recall

Recall measures how many items of a true class were captured by the cluster assigned to that class:

$$R(\text{class}_c) = \frac{\text{count of class}_c \text{ in its assigned cluster}}{\text{total items of class}_c}$$

$$\begin{aligned}
R(\text{Red}) &= \frac{4}{8} = \frac{1}{2} = 0.5000 \\
R(\text{Blue}) &= \frac{5}{10} = \frac{1}{2} = 0.5000 \\
R(\text{Purple}) &= \frac{4}{9} \approx 0.4444
\end{aligned}$$

Macro-Averaged Recall:

$$\begin{aligned}
\overline{R} &= \frac{1}{3}\left(\frac{1}{2} + \frac{1}{2} + \frac{4}{9}\right) \\
&= \frac{1}{3}\left(\frac{9}{18} + \frac{9}{18} + \frac{8}{18}\right) \\
&= \frac{1}{3} \cdot \frac{26}{18} = \frac{26}{54} = \frac{13}{27}
\end{aligned}$$

$$\overline{R} = \frac{13}{27} \approx 0.4815$$

2.4 Macro-Averaged F-score

$$\overline{F} = \frac{2\,\overline{P}\,\overline{R}}{\overline{P} + \overline{R}}$$

Step 1 — Denominator:

$$\overline{P} + \overline{R} = \frac{131}{270} + \frac{13}{27} = \frac{131}{270} + \frac{130}{270} = \frac{261}{270} = \frac{29}{30}$$

Step 2 — Numerator:

$$2\,\overline{P}\,\overline{R} = 2 \times \frac{131}{270} \times \frac{13}{27} = \frac{2 \times 131 \times 13}{270 \times 27} = \frac{3406}{7290}$$

Step 3 — Divide:

$$\overline{F} = \frac{3406/7290}{261/270} = \frac{3406}{7290} \times \frac{270}{261} = \frac{3406 \times 270}{7290 \times 261} = \frac{3406}{27 \times 261} = \frac{3406}{7047}$$

$$\overline{F} = \frac{3406}{7047} \approx 0.4833$$

Question 2 Summary

$$\overline{P} = \frac{131}{270} \approx 0.4852, \qquad \overline{R} = \frac{13}{27} \approx 0.4815, \qquad \overline{F} = \frac{3406}{7047} \approx 0.4833$$

3. Question 3 (10 pts): B-CUBED Precision, Recall, and F-score

3.1 B-CUBED Definitions

For each item e, let $L(e)$ be its *true class* and $K(e)$ be the *cluster* it is assigned to. The per-item scores are:

$$P_{\text{BC}}(e) = \frac{|\{e' \in K(e) : L(e') = L(e)\}|}{|K(e)|} \qquad \text{(cluster purity for } e\text{)}$$

$$R_{\text{BC}}(e) = \frac{|\{e' \in K(e) : L(e') = L(e)\}|}{|\{e'' : L(e'') = L(e)\}|} \qquad \text{(class coverage for } e\text{)}$$

The overall B-CUBED scores are the *averages* over all $N = 27$ items:

$$P_{\text{BC}} = \frac{1}{N}\sum_e P_{\text{BC}}(e), \qquad R_{\text{BC}} = \frac{1}{N}\sum_e R_{\text{BC}}(e)$$

3.2 Per-Item B-CUBED Precision

Items sharing the same cluster *and* class have the same precision value, so we compute once per (*class*, *cluster*) cell and multiply by the cell count.

Class	Cluster (label)	n_{ij}	$\lvert K\rvert$	$P_{\text{BC}}(e)$
Red	Red (C1)	4	10	4/10
Blue	Red (C1)	3	10	3/10
Purple	Red (C1)	3	10	3/10
Red	Blue (C2)	2	9	2/9
Blue	Blue (C2)	5	9	5/9
Purple	Blue (C2)	2	9	2/9
Red	Purple (C3)	2	8	2/8
Blue	Purple (C3)	2	8	2/8
Purple	Purple (C3)	4	8	4/8

Weighted sum of individual precisions:

$$\begin{aligned}
\sum P &= 4\cdot\tfrac{4}{10} + 3\cdot\tfrac{3}{10} + 3\cdot\tfrac{3}{10} + 2\cdot\tfrac{2}{9} + 5\cdot\tfrac{5}{9} + 2\cdot\tfrac{2}{9} + 2\cdot\tfrac{2}{8} + 2\cdot\tfrac{2}{8} + 4\cdot\tfrac{4}{8} \\
&= \frac{16+9+9}{10} + \frac{4+25+4}{9} + \frac{4+4+16}{8} \\
&= \frac{34}{10} + \frac{33}{9} + \frac{24}{8} \\
&= \frac{306}{90} + \frac{330}{90} + \frac{270}{90} = \frac{906}{90} = \frac{151}{15}
\end{aligned}$$

$$P_{\mathrm{BC}} = \frac{151/15}{27} = \frac{151}{405} \approx 0.3728$$

3.3 Per-Item B-CUBED Recall

Class	Cluster (label)	n_{ij}	$\lvert C\rvert$	$R_{\mathrm{BC}}(e)$
Red	Red (C1)	4	8	4/8
Blue	Red (C1)	3	10	3/10
Purple	Red (C1)	3	9	3/9
Red	Blue (C2)	2	8	2/8
Blue	Blue (C2)	5	10	5/10
Purple	Blue (C2)	2	9	2/9
Red	Purple (C3)	2	8	2/8
Blue	Purple (C3)	2	10	2/10
Purple	Purple (C3)	4	9	4/9

Weighted sum of individual recalls:

$$\begin{aligned}
\sum R &= 4\cdot\tfrac{4}{8} + 3\cdot\tfrac{3}{10} + 3\cdot\tfrac{3}{9} + 2\cdot\tfrac{2}{8} + 5\cdot\tfrac{5}{10} + 2\cdot\tfrac{2}{9} + 2\cdot\tfrac{2}{8} + 2\cdot\tfrac{2}{10} + 4\cdot\tfrac{4}{9} \\
&= \frac{16+4+4}{8} + \frac{9+25+4}{10} + \frac{9+4+16}{9} \\
&= \frac{24}{8} + \frac{38}{10} + \frac{29}{9}
\end{aligned}$$

Convert to a common denominator of $\mathrm{lcm}(8, 10, 9) = 360$:

$$\begin{aligned}
&= \frac{24\times 45}{360} + \frac{38\times 36}{360} + \frac{29\times 40}{360} \\
&= \frac{1080+1368+1160}{360} = \frac{3608}{360} = \frac{451}{45}
\end{aligned}$$

$$R_{\mathrm{BC}} = \frac{451/45}{27} = \frac{451}{1215} \approx 0.3712$$

3.4 B-CUBED F-score

$$F_{\mathrm{BC}} = \frac{2\,P_{\mathrm{BC}}\,R_{\mathrm{BC}}}{P_{\mathrm{BC}} + R_{\mathrm{BC}}}$$

Step 1 — Denominator:

$$P_{\mathrm{BC}} + R_{\mathrm{BC}} = \frac{151}{405} + \frac{451}{1215} = \frac{453}{1215} + \frac{451}{1215} = \frac{904}{1215}$$

Step 2 — Numerator:

$$2\,P_{\mathrm{BC}}\,R_{\mathrm{BC}} = 2 \times \frac{151}{405} \times \frac{451}{1215} = \frac{2 \times 151 \times 451}{405 \times 1215} = \frac{136\,202}{492\,075}$$

Step 3 — Divide:

$$F_{\mathrm{BC}} = \frac{136\,202\,/\,492\,075}{904\,/\,1215} = \frac{136\,202}{492\,075} \times \frac{1215}{904} = \frac{136\,202}{405 \times 904} = \frac{136\,202}{366\,120} = \frac{68\,101}{183\,060}$$

$$\boxed{F_{\mathrm{BC}} = \frac{68\,101}{183\,060} \approx 0.3720}$$

Question 3 Summary

$$P_{\mathrm{BC}} = \frac{151}{405} \approx 0.3728, \qquad R_{\mathrm{BC}} = \frac{451}{1215} \approx 0.3712, \qquad F_{\mathrm{BC}} = \frac{68\,101}{183\,060} \approx 0.3720$$

4. Combined Summary

	Precision	Recall	F-score
Q2: Macro-Averaged	$\frac{131}{270} \approx 0.4852$	$\frac{13}{27} \approx 0.4815$	$\frac{3406}{7047} \approx 0.4833$
Q3: B-CUBED	$\frac{151}{405} \approx 0.3728$	$\frac{451}{1215} \approx 0.3712$	$\frac{68101}{183060} \approx 0.3720$